Many areas of machine learning focus on improving performance using increasingly large datasets and established benchmarks. Emerging biological measurement technologies present a different challenge. Before models can be optimised, researchers must first determine what information the data contains, how it should be represented, and which signals are biologically meaningful. RNA fragmentomics is particularly interesting in this regard. The combination of quantitative and positional information creates opportunities for entirely new biomarkers, but also raises methodological questions that do not yet have established answers. That combination of methodological development and clinical relevance is what attracted me to this Industrial PhD position.

I have recently completed my Master's degree in Data Science at Uppsala University, where I gradually focused my interests on healthcare and machine learning. What initially drew me to the field was how data-driven systems can improve healthcare while remaining trustworthy and useful in practice. Early on, I wrote about bias in medical AI systems and how existing limitations in healthcare research risk being amplified as increasingly capable models are deployed. Since then, I have become increasingly interested in how data quality, representation, and modelling choices influence the conclusions we draw from biomedical data.

Through a health innovation course, I studied the European health technology ecosystem and the challenges involved in bringing new technologies into clinical use. In another project, I investigated healthcare data interoperability challenges in Sweden, examining how information moves between systems and organisations. More recently, I started a pilot collaboration with the Haematology and Oncology Department at King Faisal Specialist Hospital in Saudi Arabia, exploring AI-assisted interpretation of electrophoresis results. While these experiences differ in scope and technical depth, they all reinforced the same observation: healthcare data is rich, fragmented, and often provides only a partial view of the underlying biological processes.

My strongest technical experience comes from machine learning and software development. My master's thesis focuses on multimodal machine learning for harmful content detection, combining text, image, and contextual information within a unified modelling framework. The work required developing custom PyTorch architectures, designing representation learning approaches, handling large datasets, and making modelling decisions under significant ambiguity. Different domain, but key skills are transferable: integrating heterogeneous data sources, learning useful representations from incomplete information, and identifying which information is genuinely predictive versus merely correlated.

What particularly attracts me to this position is the opportunity to contribute to a field that is still being defined. Developing methods for RNA fragmentomics requires both statistical thinking and practical machine learning, while remaining closely connected to biological and clinical questions. The industrial PhD structure is also appealing because it allows research questions to be explored in close proximity to their eventual application. Working across MultiD, the University of Gothenburg, and healthcare collaborators provides an opportunity to understand not only how methods are developed, but also how they are evaluated and translated into tools that may ultimately influence patient care.